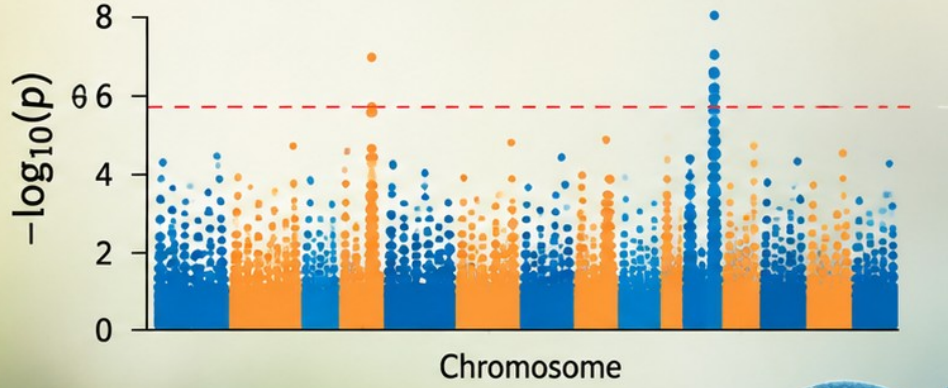
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

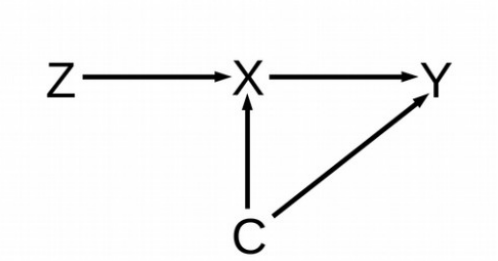
Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



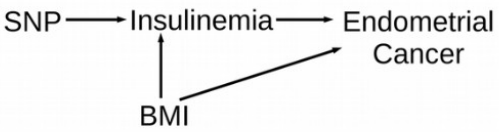


Causality Inference: get rid of confounders

$$Y \sim X + C, \text{ where C is a confounder}$$



Mendelian Randomization



Method of Instrumental Variables

```
N <- 1e3 # Number of Observations
C <- rnorm(n = N); X <- 2 * C + rnorm(n = N)
Y <- 2 * X + 2 * C + rnorm(n = N)
summary(lm(Y ~ X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.4905 -0.8991  0.0072  0.9051  4.8766
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.04544    0.04245    1.07    0.285
## X            2.79987    0.01876  149.27   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.339 on 998 degrees of freedom
## Multiple R-squared:  0.9571,    Adjusted R-squared:  0.9571
## F-statistic: 2.228e+04 on 1 and 998 DF,  p-value: < 2.2e-16
```

Known confounders: Controlled Regression

```
summary(lm(Y ~ X + C))
```

```
##
## Call:
## lm(formula = Y ~ X + C)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1141 -0.6955 -0.0089  0.7270  3.8172
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.06693    0.03250    2.06   0.0397 *
## X            2.00552    0.03314   60.52   <2e-16 ***
## C            1.97305    0.07419   26.59   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.025 on 997 degrees of freedom
## Multiple R-squared:  0.9749,    Adjusted R-squared:  0.9749
## F-statistic: 1.938e+04 on 2 and 997 DF,  p-value: < 2.2e-16
```

What if confounders are not known?

Good Instrumental Variable

```r
set.seed(1)
library("AER")
N <- 1e3 # Number of Observations
weak <- rnorm(n = N)
good <- rnorm(n = N)
C <- rnorm(n = N)
X <- 0.000001 * weak + 2 * good + 2 * C + rnorm(n = N)
Y <- 2 * X + 2 * C + rnorm(n = N)
df <- cbind(Y, X, C, good, weak)
colnames(df) <- c('Y', 'X','C', 'good', 'weak')
df <- data.frame(df)
summary(ivreg(Y ~ X | good, data = df))
```

```
##
## Call:
## ivreg(formula = Y ~ X | good, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.78427 -1.48164  0.05357  1.37862  6.99009
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01017    0.06867   0.148    0.882
## X            2.02609    0.03207  63.183   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.172 on 998 degrees of freedom
## Multiple R-Squared: 0.9237,    Adjusted R-squared: 0.9236
## Wald test:  3992 on 1 and 998 DF,  p-value: < 2.2e-16
```

Weak Instrumental Variable

```
summary(ivreg(Y ~ X | weak, data = df))
```

```
##
## Call:
## ivreg(formula = Y ~ X | weak, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.319 -1.157  0.027  1.218  5.711
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.003767   0.056301   0.067    0.947
## X           2.458141   0.401503   6.122 1.32e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.77 on 998 degrees of freedom
## Multiple R-Squared: 0.9493,    Adjusted R-squared: 0.9492
## Wald test: 37.48 on 1 and 998 DF,  p-value: 1.324e-09
```

What does ivreg do under the hood?

```
summary(lm(Y~as.numeric(predict(lm(X~good)))))
```

```
##
## Call:
## lm(formula = Y ~ as.numeric(predict(lm(X ~ good))))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.8345  -4.2676  -0.0506   4.3873  20.1810
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     0.01017    0.20723   0.049    0.961
## as.numeric(predict(lm(X ~ good)))  2.02609    0.09677  20.938   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.553 on 998 degrees of freedom
## Multiple R-squared:  0.3052,    Adjusted R-squared:  0.3045
## F-statistic: 438.4 on 1 and 998 DF,  p-value: < 2.2e-16
```

Lasso helps to build good Instrumental Variables

